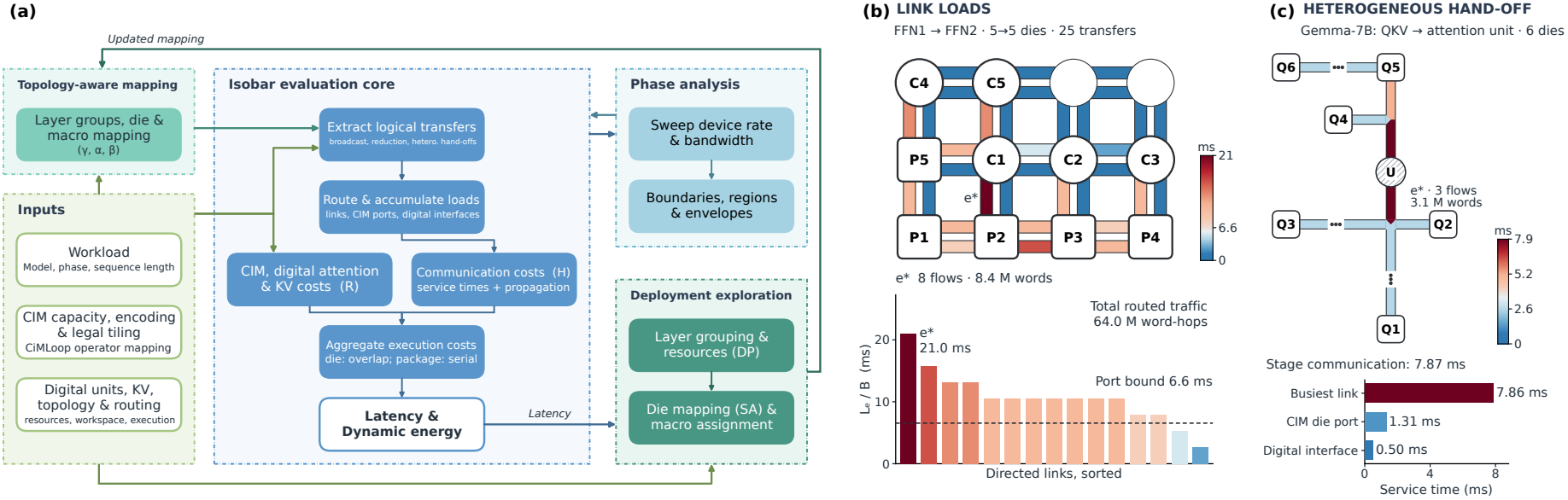
(a)
Updated mapping
Topology-aware mapping
Layer groups, die & macro mapping
(γ, α, β)
Inputs
Workload
Model, phase, sequence length
CIM capacity, encoding & legal tiling
CiMLoop operator mapping
Digital units, KV, topology & routing
resources, workspace, execution
Isobar evaluation core
Extract logical transfers
broadcast, reduction, hetero. hand-offs
Route & accumulate loads
links, CIM ports, digital interfaces
CIM, digital attention & KV costs (R)
Communication costs (H)
service times + propagation
Aggregate execution costs
die: overlap; package: serial
Latency & Dynamic energy
Latency
Phase analysis
Sweep device rate & bandwidth
Boundaries, regions & envelopes
Deployment exploration
Layer grouping & resources (DP)
Die mapping (SA) & macro assignment
(b) LINK LOADS
FFN1 → FFN2 · 5→5 dies · 25 transfers
C4
C5
P5
C1
C2
C3
e*
P1
P2
P3
P4
ms
21
6.6
0
e* 8 flows · 8.4 M words
Total routed traffic
64.0 M word-hops
e*
21.0 ms
Port bound 6.6 ms
Le / B (ms)
20
10
0
Directed links, sorted
(c) HETEROGENEOUS HAND-OFF
Gemma-7B: QKV → attention unit · 6 dies
Q6
Q5
Q4
U
e* · 3 flows
3.1 M words
Q3
Q2
Q1
ms
7.9
5.2
2.6
0
Stage communication: 7.87 ms
Busiest link
7.86 ms
CIM die port
1.31 ms
Digital interface
0.50 ms
0
4
8
Service time (ms)